Semantic Exploration Analysis

Analysis of Semantic Exploration Bins

PENPAL Project

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0.1 Summary

- Semantic distances decrease with larger bin sizes, reflecting expected averaging across longer interaction windows.
- In the field data, AI contributions show slightly higher centroid distances than user contributions, indicating somewhat greater semantic spread.
- No significant user–AI interaction: both follow similar decay rates as bin size increases.
- Starter identity (user vs AI) has minimal effect on exploration patterns.
- In interleaved AI–AI runs, both models show similar behaviour, with one variant displaying faster semantic compression.
- Overall, exploration patterns are structured and nonrandom, with AI in the field dataset often making the larger semantic jumps.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

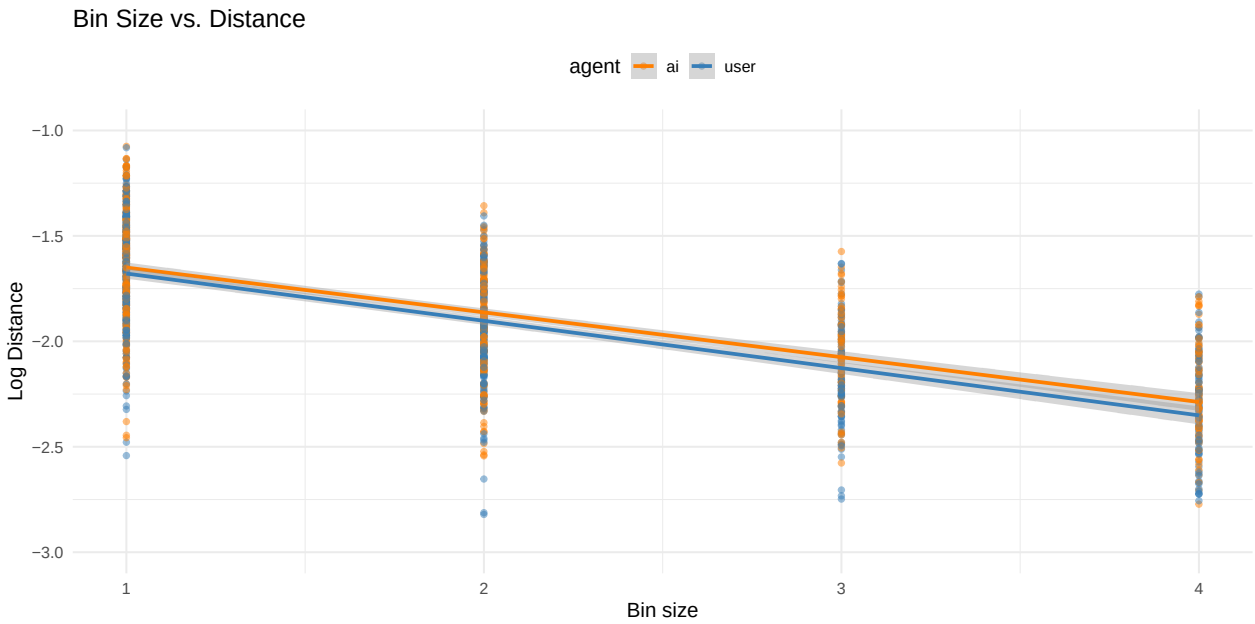
1.2 1.2 Descriptive Statistics

Table 1: Descriptive Statistics for Semantic Exploration by Agent

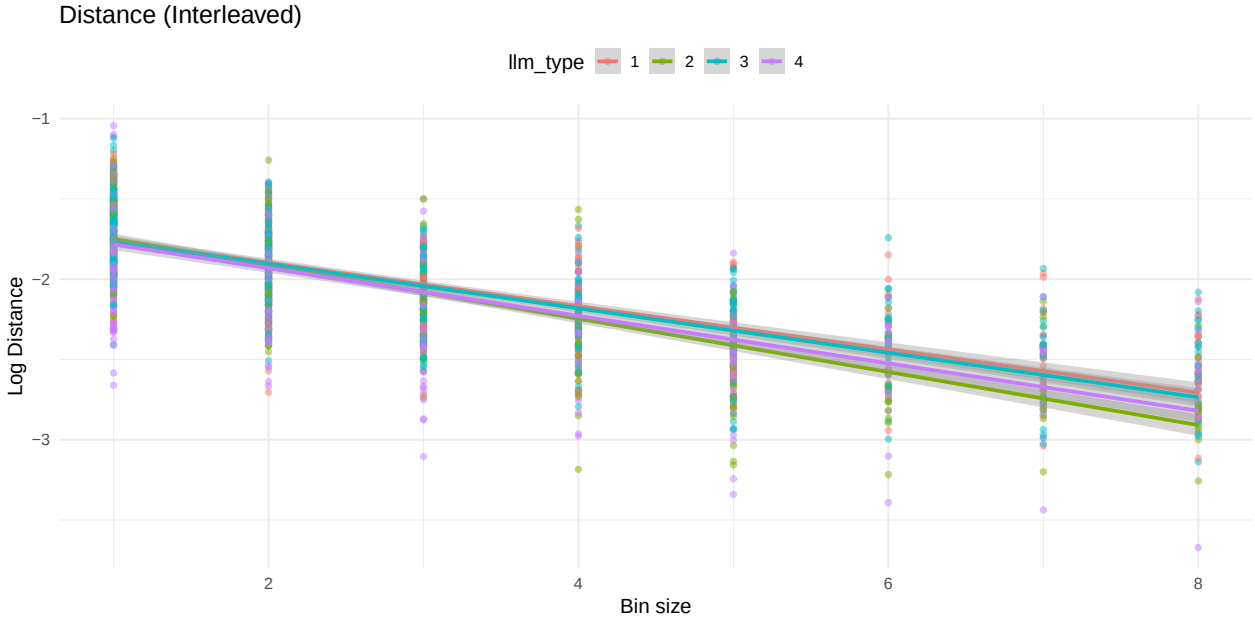
agent	n	mean_distance	sd_distance	mean_log_distance	sd_log_distance
ai	656	0.168	0.054	-1.839	0.337
interleaved	1886	0.140	0.053	-2.040	0.401
user	656	0.160	0.055	-2.104	2.745

2 2. Semantic Exploration Analysis

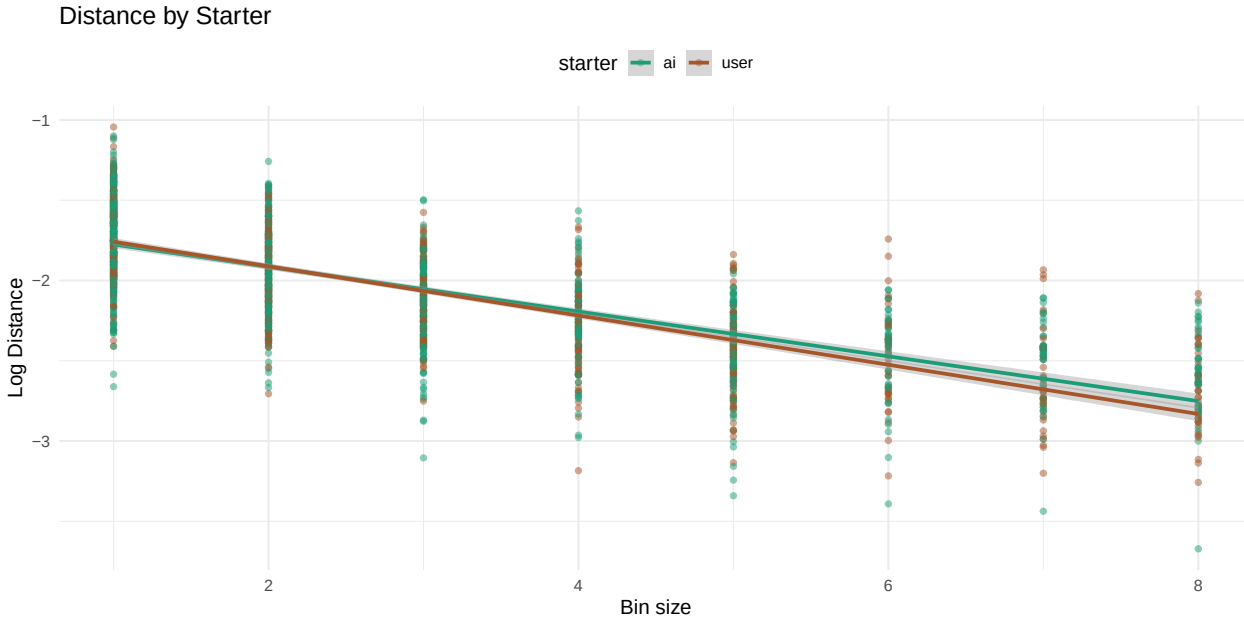
2.1 2.1 Bin Size vs. Distance Plot



2.2 2.2 Interleaved Distance Plot



2.3 2.3 Starter Effect Plot



3 3. Statistical Models

3.1 3.1 Agent Interaction Model

```
## === Agent Interaction Model ===

##                Estimate Std. Error        df   t value     Pr(>|t|)
## (Intercept) -1.44507003 0.17198973  357.6549 -8.402072 1.047754e-15
## k           -0.21012237 0.06294509 1227.0000 -3.338185 8.685695e-04
## agentuser   -0.44263929 0.19144002 1227.0000 -2.312156 2.093396e-02
## k:agentuser  0.09468189 0.08901780 1227.0000  1.063629 2.877062e-01
```

3.2 3.2 LLM Type Interaction Model (Interleaved)

```
## === LLM Type Interaction Model (Interleaved) ===

##                          Estimate  Std. Error        df       t value
## (Intercept)         -1.6288618696 0.041002134  103.8472 -39.726270374
## k                   -0.1348100672 0.005226298 1800.0000 -25.794559780
## factor(llm_type)2    0.0446551038 0.057985774  103.8472   0.770104468
## factor(llm_type)3    0.0003557716 0.054240725  103.8472   0.006559123
## factor(llm_type)4   -0.0071029283 0.055287266  103.8472  -0.128473133
## k:factor(llm_type)2 -0.0307880155 0.007391102 1800.0000  -4.165551359
## k:factor(llm_type)3 -0.0036058802 0.006913743 1800.0000  -0.521552540
## k:factor(llm_type)4 -0.0132520718 0.007047139 1800.0000  -1.880489529
##                          Pr(>|t|)
## (Intercept)          1.282167e-64
## k                   4.083743e-125
## factor(llm_type)2    4.429863e-01
## factor(llm_type)3    9.947792e-01
## factor(llm_type)4    8.980230e-01
## k:factor(llm_type)2  3.253047e-05
## k:factor(llm_type)3  6.020461e-01
```

```
## k:factor(llm_type)4  6.020273e-02
```

3.3 3.3 Starter Interaction Model (Interleaved)

```
## === Starter Interaction Model (Interleaved) ===

##                          Estimate  Std. Error        df     t value
## (Intercept)           -1.63614874 0.026730926  106.9921 -61.2080823
## k                     -0.13938288 0.003432362 1802.0000 -40.6084377
## factor(starter)user    0.03133973 0.038272862  106.9921   0.8188498
## k:factor(starter)user -0.01393252 0.004914395 1802.0000  -2.8350418
##                            Pr(>|t|)
## (Intercept)            4.226179e-85
## k                     1.498395e-256
## factor(starter)user    4.146923e-01
## k:factor(starter)user  4.633089e-03
```

4 4. Session Information

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sequoia 15.6.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Copenhagen
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_2.0.0   here_1.0.2      afex_1.5-0      cowplot_1.2.0
##  [5] patchwork_1.3.2 zoo_1.8-14      stringi_1.8.7   arrow_22.0.0
##  [9] lmerTest_3.1-3  lme4_1.1-37     Matrix_1.7-1    lubridate_1.9.4
## [13] forcats_1.0.1   stringr_1.6.0   dplyr_1.1.4     purrr_1.2.0
## [17] readr_2.1.6     tidyr_1.3.1     tibble_3.3.0    ggplot2_4.0.1
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        S7_0.2.1
##  [4] fastmap_1.2.0       TH.data_1.1-2       pacman_0.5.1
##  [7] digest_0.6.39       timechange_0.3.0    estimability_1.5.1
## [10] lifecycle_1.0.4     survival_3.7-0      magrittr_2.0.4
## [13] compiler_4.4.2      rlang_1.1.6         tools_4.4.2
## [16] yaml_2.3.10         knitr_1.50          labeling_0.4.3
## [19] bit_4.6.0           plyr_1.8.9          RColorBrewer_1.1-3
## [22] abind_1.4-8         multcomp_1.4-26     withr_3.0.2
## [25] numDeriv_2016.8-1.1 grid_4.4.2          xtable_1.8-4
```

```
## [28] scales_1.4.0        MASS_7.3-61         tinytex_0.58
## [31] cli_3.6.5           mvtnorm_1.3-3       rmarkdown_2.30
## [34] ragg_1.5.0          reformulas_0.4.2    generics_0.1.4
## [37] rstudioapi_0.17.1   reshape2_1.4.5      tzdb_0.5.0
## [40] minqa_1.2.8         splines_4.4.2       assertthat_0.2.1
## [43] parallel_4.4.2      vctrs_0.6.5         boot_1.3-31
## [46] sandwich_3.1-1      carData_3.0-5       car_3.1-3
## [49] hms_1.1.4           bit64_4.6.0-1       Formula_1.2-5
## [52] systemfonts_1.3.1   glue_1.8.0          nloptr_2.2.1
## [55] codetools_0.2-20    gtable_0.3.6        pillar_1.11.1
## [58] htmltools_0.5.8.1   R6_2.6.1            textshaping_1.0.4
## [61] Rdpack_2.6.4        rprojroot_2.1.1     evaluate_1.0.5
## [64] lattice_0.22-6      rbibutils_2.4       Rcpp_1.1.0
## [67] coda_0.19-4.1       nlme_3.1-166        mgcv_1.9-1
## [70] xfun_0.54           pkgconfig_2.0.3
```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.